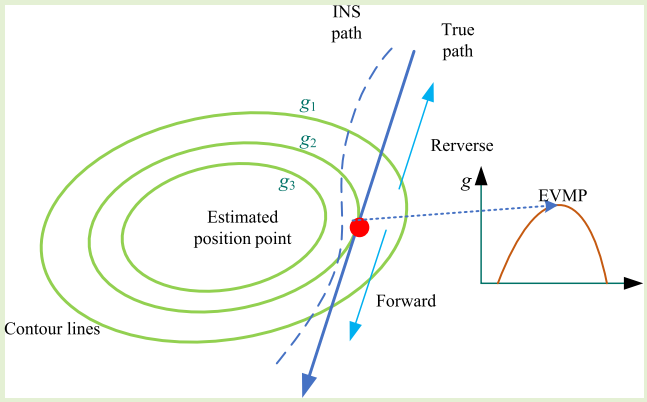
An Improved Particle Filter Based on Gravity Measurement Feature in Gravity-Aided Inertial Navigation System

Shengwu Zhao, Xuan Xiao, Yu Wang, and Zhihong Deng

***Abstract*—The existing gravity matching algorithms are affected by the initial position error of the inertial navigation system (INS), the gravity measurement error, and the similarity of the gravity background map. Aiming at the above problems, an improved particle filter based on the gravity measurement feature (IPFBGMF) is proposed in this article. In the IPFBGMF, both the value and change characteristic of gravity measurements are considered, and a novel position acquisition method based on the gravity measurement feature is proposed, which can reduce the influence of the initial position error of INS. In addition, a new concept called direction measurement using the heading angle of INS is proposed to optimize the weight of particles in the PF. The PF with direction measurement can reduce the influence of the gravity measurement error and the similarity of the gravity background map. Furthermore, the robustness of the improved PF with the precise position is proven. Finally, a navigation strategy is designed to apply the proposed algorithms. Simulations show that IPFBGMF has the highest positioning accuracy compared with the traditional gravity matching algorithms.**



***Index Terms*—Extreme value, gravity-aided inertial navigation system (GAINS), gravity measurement feature, particle filter (PF), underwater navigation.**

I. Introduction

SINCE 21st century, most countries have taken ocean exploration and ocean resource exploitation as an important strategy. An underwater vehicle is a piece of indispensable carrier equipment for ocean exploration and development, which is widely used in ocean surveys, ocean rescue, and military fields. Autonomous underwater navigation technology is the key technology of underwater vehicles and has become the research hotspot of researchers all over the world [1], [2]. Inertial navigation system (INS) has become the main navigation mode in the field of underwater navigation due to its characteristics of complete autonomy, high concealment, no environmental influence, and comprehensive navigation information. However, the errors of INS accumulate with time, which cannot meet the requirements of positioning accuracy of underwater vehicles for long time operation [3]. Therefore, other navigation methods are needed to compensate for the errors of INS.

The ocean gravity field is stable and does not change with time. Thus, the gravity-aided INS (GAINS) becomes an effective way of underwater navigation [4]. The process of GAINS is shown in Fig. 1. Among them, gravity matching is the core [5]. It is used to obtain the position estimation of underwater vehicles by integrating the output data of INS, the prestored background map of the gravity field, and the gravity anomaly measurements in real time, providing the basis for the position readjustment of INS comprehensive correction. The gravity-matching algorithms can be divided into two categories: sequence-matching algorithms and single-point algorithms.

Sequence-matching algorithms include iterated closest contour point (ICCP), terrain contour matching (TERCOM), and related improved algorithms. Mahalanobis distance in [6] and Hausdorff distance in [7] instead of Euclidean distance

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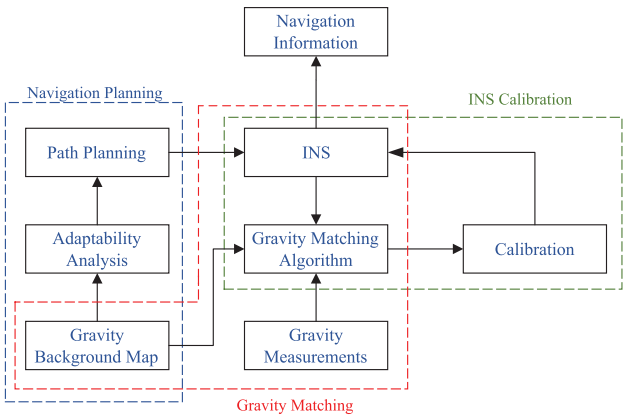

Fig. 1. Framework of GAINS.

are utilized to improve the accuracy of ICCP. A method combining Sage–Husa adaptive filtering and a strong-tracked Kalman filter [8] is used to improve the performance of ICCP [9]. In [10], the shortest path algorithm is applied to TERCOM, and a novel correlation analysis method is also proposed. A mismatch diagnostic method based on the restricted spatial-order constraints for TERCOM in GAINS is proposed in [11]. In addition, a novel gravity-matching navigation algorithm based on multiscale search and Hadamard transformed difference is proposed in [12]. Li et al. [13] used an a priori recursive iterative least-squares mismatching correction method for gravity matching. In [14], a triangle-matching algorithm is proposed to bound the drift errors in GAINS. A novel gravity-matching algorithm based on the K-nearest neighbor is proposed in [15].

Most single-matching algorithms are real-time, while sequence-matching algorithms are not. It includes Sandia terrain-aided navigation (SITAN), point mass filter (PMF), particle filter (PF), and related algorithms. Dai and Kang [16] combined the contour information by the method of interpolation and linear processing and used SITAN to realize geomagnetic matching. In [17], TERCOM is used to reduce initial error, and then the optimal estimation position can be obtained by SITAN. Other filter algorithms can also be applied in GAINS. For instance, unscented Kalman filter (UKF) [18] and cubature Kalman filter (CKF) [19]. In [20], PMF is first used to GAINS. Then, Han et al. [21] improved the PMF with similarity transformation and proposed a two-stage matching process. A vector-matching method based on PF is proposed in [22] which uses the shape of INS, and later it is improved in [23] since the range of particles is considered. Some intelligent algorithms can be single-point-matching algorithms, such as artificial bee colony [24] and particle swarm optimization [25]. Apart from improved sequence or single-point-matching algorithms, these algorithms can also be combined. For instance, Yuan et al. [17] combined TERCOM and SITAN, and Han et al. [26] combined ICCP and PMF.

However, the existing gravity-matching algorithms are affected by INS initial position error, gravity measurement error, and the similarity of gravity background map. In this article, we consider the change characteristic of the gravity anomaly values rather than just the value, on the basis of which an accurate position point can be obtained. In addition, a novel concept called direction measurement utilizing the heading angle of INS is proposed in view of the similarity of the gravity background map. Finally, the navigation strategy is given. To be specific, the main contributions of this work can be concluded as follows.

1) A novel position acquisition method is proposed, which considers the gravity measurement feature including the value and the change characteristic. The sliding window method is designed to find the extreme point in the measurement sequence within a period of time. The tangent direction of the points on the contour line corresponding to the extreme point is calculated. Combined with the heading angle of INS, the precision position point can be obtained by the weighting method, which can reduce the influence of the INS initial position error.
2) The defects of PF applied to INS are analyzed, and a novel direction measurement is proposed, which uses the INS heading angle at the intermediate moment of the matching interval. The direction measurement is seen as a virtual measurement and added to PF. The improved PF is not sensitive to gravity measurement error and the similarity of the gravity background map. In addition, the improved PF with the accurate position point is robust to outliers, which is proven.
3) A new matching strategy is designed, which provides the flow of underwater vehicles applying the above-proposed algorithms in the adaptation area.

The rest of this article is arranged as follows: some research on gravity measurement features and how to obtain precise estimated position points are in Section II. In Section III, direction measurement is proposed, and PF is improved. Improved PF based on gravity measurement feature (IPFBGMF) is designed in Section IV. Simulation and analysis are in Section V. Section VI is the conclusion.

II. Novel Position Acquisition Method Based on Gravity Measurement Feature

Most of the traditional gravity-matching algorithms only consider the value of the gravity anomaly measurement data without analyzing the change characteristics of the data. This section proposes a novel position acquisition method based on the gravity measurement feature. The method analyzes the change characteristics of the gravity anomaly measurement data and further obtains the estimated position point by studying the extreme points of the gravity anomaly measurement data.

First, five trajectories in the adaptation area are taken as an example, which are shown in Fig. 2(a). The black lines represent the contour line corresponding to the gravity anomaly value $g_i (i = 1, 2, 3, 4, 5)$. Set $g_1 > g_2 > g_3 > g_4 > g_5$, then the changing trend of gravity anomaly values measured along the five types of paths in Fig. 2(a) is shown in Fig. 2(b).

Next, focus on the changing trend of gravity anomaly values on Traj_2. The measured gravity anomaly values change from small to large to small, reaching a local maximum at time t.

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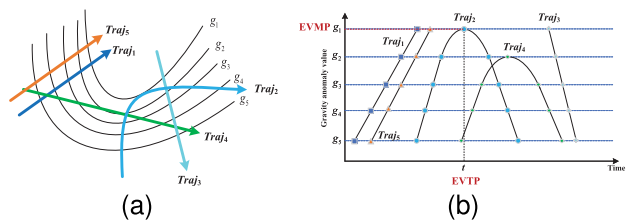

Fig. 2. (a) Five types of underwater vehicle paths. (b) Gravity anomaly values corresponding to five types of underwater vehicle paths.

The corresponding gravity anomaly measurement value is g_1 at this time. The time point t is recorded as the extreme value time point (EVTP), and g_1 is recorded as the extreme value measurement point (EVMP).

It can be seen that the gravity anomaly measurement value sequence corresponding to Traj_4 also has EVMP. Fig. 2(a) shows that the path of Traj_2 and Traj_4 are both tangent to a certain contour line. The gravity anomaly value corresponding to the tangent contour is the EVMP. The tangential angle is the current heading angle of the underwater vehicle, and the tangent point is the current position of the underwater vehicle. When two curves are tangent, in order to find the tangent point, it is necessary to find the corresponding tangential angle. In the absence of input of external information, the available angle information only comes from the heading angle output by INS. Therefore, the main work of this section includes finding the EVMPs and EVTPs of the real-time gravity measurement value sequence, and obtaining the underwater vehicle's estimated position point at EVTP through the output heading angle of INS and the contour corresponding to the EVMP.

A. Generation of EVMP and EVTP

As shown above, the extreme points of gravity anomaly measurement values need to be studied. After the underwater vehicle enters the adaptation area, the carrier direction is adjusted according to the changes in real-time gravity anomaly measurement value, until an extreme point appears within a certain period of time. The path-planning algorithm [27] can be used to make the path tangent to a contour line.

B. Determination of EVMP and EVTP

When an extreme point occurs within a certain time period, it is necessary to find it. For the gravity anomaly measurement value sequence without measurement error, the extreme point can be obtained by fitting the local function and then making the first derivative zero. However, the actual gravity anomaly measurement value sequence inevitably has errors. The function fitting method may cause multiple extreme points in a local area, and the actual EVMPs may not be the highest or lowest point in the local area due to the interference of measurement errors. Through the analysis of Fig. 2(a) and (b), it can be obtained that the change degree of the gravity anomaly measurement value near EVTP is slower than that at other time points. Therefore, a sliding window with a length of N (N is an odd number) is designed in this section, which can

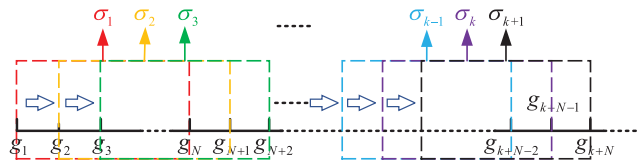

Fig. 3. Sliding window for getting extreme value point.

obtain the extreme value point by judging the change degree of the gravity anomaly measurement value. It can be shown in Fig. 3.

At time k, for the previous N consecutive gravity anomaly values, calculate the standard deviation of these data σ_{k-N+1} by

$$\sigma_{k-N+1} = \sqrt{\frac{1}{N-1}\sum_{i=k-N+1}^{k}(g_i - \bar{g}_{k-N+1})^2} \tag{1}$$

and

$$\bar{g}_{k-N+1} = \frac{1}{N}\sum_{i=k-N+1}^{k} g_i \tag{2}$$

where g_i is the gravity anomaly value at time i. In the area where extreme points may appear, calculate the standard deviation of the data in each window, and find the sliding window corresponding to the minimum of these standard deviations. Use the data at the center of the sliding window as EVMP. EVTP is determined as the time of the EVMP. The value of N needs to ensure that the extreme point can be clearly distinguished in this window. The value of N is affected by the measurement error and needs to be determined according to the matching period of the actual voyage. The larger the measurement error, the larger the value of N should be. When the matching period is too large, N should be appropriately reduced to improve the computational efficiency. Through many simulations, $N \in [7, 15]$.

C. Determination of Tangential Angle

When two curves are tangent, the tangential angle is needed to determine the tangent point. This section uses the heading angle output by INS to determine the tangential angle. Generally speaking, the output frequency of INS is greater than the frequency of gravity measurement. Therefore, when EVMP and EVTP are determined, the heading angle φ output by INS at this time can be used as the tangential angle.

D. Determination of Estimated Position

This section focuses on the contour line corresponding to EVMP obtained earlier. The contour line calculated by MATLAB is expressed in the form of multiple position points, which have the same gravity anomaly value. When position points on a contour line are obtained, they should be arranged clockwise or counterclockwise.

First, calculate the tangent direction of each point on the contour line. The tangent direction of each point is calculated by approximating the contour line locally as a circular arc.

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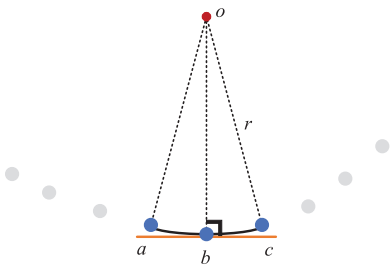

Fig. 4. Schematic of calculation of tangent direction.

Take three continuous discrete points $a(x_1, y_1)$, $b(x_2, y_2)$, and $c(x_3, y_3)$ as an example. First, calculate the radius and center of the circle composed of these three points. As shown in Fig. 4, the tangent direction of point b can be calculated by the following equation:

$$k_b = -\frac{x_c - x_2}{y_c - y_2} \tag{3}$$

where (x_c, y_c) is the center of the circle.

By performing the above operation on all points on the contour line, the tangent direction of all points can be obtained. Since the contour is approximately continuous, the change in the tangent direction is also approximately continuous. Make $k = \tan\varphi$, in which φ is the head angle output by INS in Section II-C. When k is between the tangent direction of two consecutive points, for instance, $k_m < k < k_n$, one possible estimated position point $(\hat{x}, \hat{y})$ can be got by the following equations:

$$\hat{x} = \frac{\varphi - \varphi_m}{\varphi_n - \varphi_m} x_n + \frac{\varphi_n - \varphi}{\varphi_n - \varphi_m} x_m \tag{4}$$

$$\hat{y} = \frac{\varphi - \varphi_m}{\varphi_n - \varphi_m} y_n + \frac{\varphi_n - \varphi}{\varphi_n - \varphi_m} y_m \tag{5}$$

where $m(x_m, y_m)$ and $n(x_n, y_n)$ are the points corresponding to tangent direction k_m and k_n, $\varphi_m = \arctan k_m$, and $\varphi_n = \arctan k_n$.

Obviously, the estimated position point may be more than one, so it needs to be determined according to the INS path and the navigation time. A contour line in the gravity background map generally has a wide distribution range, and the distance between several estimated position points obtained is relatively far. When the navigation time is not long, or INS has been readjusted before, at this time, the estimated position point can be judged by the INS path. The point closest to the INS path is selected as the position of the current EVTP.

In summary, the flowchart is shown in Fig. 5.

E. Factors Affecting the Estimation Accuracy of Estimated Position Point

The position estimation points can be obtained through Sections II-B–II-D. The factors that affect the estimation accuracy include the construction error of the gravity background map, the contour line construction error, the gravity anomaly measurement error, INS output heading angle error, the calculation error, and so on. These errors can be reduced by improving the algorithm and device precision.

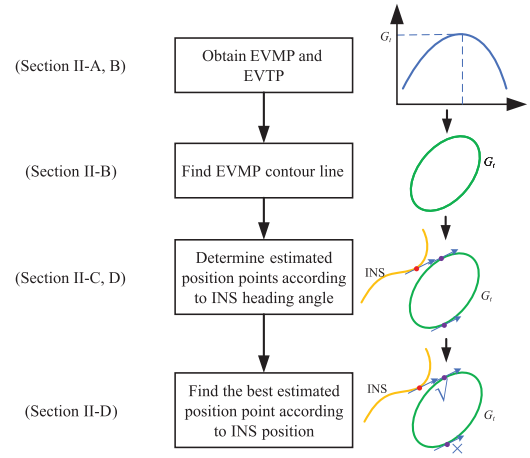

Fig. 5. Flowchart of determination of estimated position point.

However, when the tangent direction of the point on a certain segment of the contour line does not change much, the error of the estimated position point may become larger under the influence of the above error. Therefore, the path planning algorithm needs to consider this problem, which will be our future work.

In summary, an accurate estimated position point can be obtained through this section. Next, a proper gravity matching algorithm should be chosen to use the estimated position point. Kalman filter algorithms (SITAN, UKF, CKF) need more information, while only position information can be estimated. ICCP is affected by gravity measurement error. PMF has a large computational burden although it is not sensitive to measurement error. Compared to PMF, PF is efficient in GAINS and also robust to measurement error and outliers.

Hence, the estimated position point is set as the initial position point of the PF to speed up the convergence. The whole path in the adaptation area is smoothed and filtered through reverse and forward filters in Section III.

III. Improved PF

In 1993, British scholars, Salmond and Gordon, proposed PF, which is characterized by a nonparametric form of Monte Carlo simulation [28], and can solve the state estimation problem of nonlinear systems with non-Gaussian noise characteristics. PF is a sequential Monte Carlo simulation method based on the Bayesian estimation principle, which uses a group of random particles to simulate the posterior probability distribution of the state.

A. PF in INS

In INS, in order to reduce the computational burden, PF usually uses the position model which is

$$x_k = x_{k-1} + u_{k-1,k} + w_{k-1} \tag{6}$$

$$y_k = h(x_k) + v_k \tag{7}$$

where x_k represents the position of the underwater vehicle at time k (defined in this article as a 2-D variable

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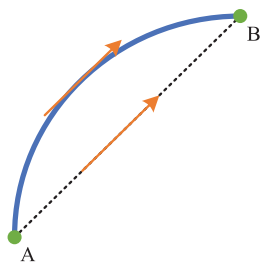

Fig. 6. Direction measurement schematic.

[latitude, longitude]), $u_{k-1,k}$ represents the displacement given by INS from time $k-1$ to time k, w_{k-1} represents INS error, y_k represents the gravity anomaly measurement value obtained by gravimeter, $h(\cdot)$ represents the process of reading out the gravity anomaly value from the gravity background map according to the position, and v_k represents the measurement error.

According to the principle of PF, when weighing, particles distributed near the contour line have a larger weight since the gravity anomaly value corresponds to a contour line in the gravity background map. More particles have similar weights, which makes PF difficult to converge to the correct position after resampling. It is caused by the similarity of the gravity field. The similarity is reflected in the fact that even with continuous gravity anomaly measurement values, it is still difficult to find a unique sequence of positions corresponding to them in the gravity background map. Not to mention that in the PF algorithm, even with gravity anomaly measurement value, there are still many similar points on the contour line.

Obviously, only gravity anomaly measurement information is not enough. Since the external measurement information available in GAINS is only the measurement information from the gravimeter, this article uses the heading angle information output by INS to construct the measurement information about the direction.

B. Direction Measurement

Assuming that the interval between two adjacent gravity matching is T. As shown in Fig. 6, the thick blue line represents the path from point A to point B sailed by the underwater vehicle within T time, and the black dotted line represents the displacement during this period of time. If the underwater vehicle does not make a sharp turn during this period of time, which causes the motion state (including heading and speed) to change significantly, the direction of the displacement during this period is roughly the same as the heading at the middle of the period. Fig. 6 shows the relationship between the displacement direction and the heading in the middle time when the path is an arc. Obviously, the above holds true when the path is a straight line. The heading angle error output by INS is in the form of oscillation and will not diverge with time. Therefore, in this article, the heading angle at the intermediate moment is constructed as the direction measurement. Together with the gravity anomaly measurement, it is used as the measurement information of PF, and the posterior weight of the particle is comprehensively calculated. Then the measurement equation of PF becomes

$$z_k = \eta(x_k) + \varepsilon_k \tag{8}$$

where $z_k = [y_k, r_k]$, $\eta(x_k) = [h(x_k), q(x_k)]^T$, and $\varepsilon_k = [v_k, \xi_k]^T$. r_k represents the heading angle output by INS in the middle period from $k-1$ to k. $q(x_k)$ represents the direction measurement calculated from the current position x_k and the previous moment position x_{k-1}. And ξ_k represents the error.

To sum up, the new position model is shown as

$$x_k = x_{k-1} + u_{k-1,k} + w_{k-1} \tag{9}$$

$$z_k = \eta(x_k) + \varepsilon_k. \tag{10}$$

C. Improved PF in INS

PF is improved by adding direction measurement to PF. The process of improved PF is as follows.

1) For $i = 1, 2, \ldots, N$ when $k = 0$

$$x_0^{(i)} \sim p_0(x_0) \tag{11}$$

$$\omega_0^{(i)} = p_y\left(y_0 | x_0^{(i)}\right). \tag{12}$$

Normalization

$$\omega_0^{(i)} = \frac{\omega_0^{(i)}}{\Sigma_{i=1}^N \omega_0^{(i)}}. \tag{13}$$

2) For time $k (k \geq 1)$, when posterior particle $\{x_{k-1}^{(i)}\}_{i=1}^N$, posterior weights $\{\omega_{k-1}^{(i)}\}_{i=1}^N$ and posterior estimation $\hat{x}_{k-1}$ at time $k-1$ is known, then

$$x_k^{(i)} = p\left(x_k | x_{k-1}^{(i)}\right) \tag{14}$$

$$\omega_k^{(i)} = \omega_{k-1}^{(i)} p_z\left(z_k | x_k^{(i)}\right). \tag{15}$$

Normalization

$$\omega_k^{(i)} = \frac{\omega_k^{(i)}}{\Sigma_{i=1}^N \omega_k^{(i)}}. \tag{16}$$

Resampling

$$\left(\left\{x_k^{(i)}\right\}_{i=1}^N, \left\{\omega_k^{(i)}\right\}_{i=1}^N\right) = \text{Resample}\left(\left\{x_k^{(i)}\right\}_{i=1}^N, \left\{\omega_k^{(i)}\right\}_{i=1}^N\right). \tag{17}$$

Therefore, the estimation result at time k is

$$\hat{x}_k = \frac{1}{N}\sum_{i=1}^N x_k^{(i)}. \tag{18}$$

In (11), $x_0^{(i)}$ represents the ith particle in initial generation, and $\omega_0^{(i)}$ represents the weight. $p_0(x_0)$ is set as a Gaussian density function. The mean is the initial position estimation $\hat{x}_0$ which is the estimated position point obtained in Section II, and the covariance is the diagonal element of the initial covariance matrix $\mathbf{P}_0$.

In (12)

$$p_y\left(y_0 | x_0^{(i)}\right) = \frac{1}{\sigma_y\sqrt{2\pi}} \exp\left(-\frac{\left(y_0 - h\left(x_0^{(i)}\right)\right)^2}{2\sigma_y^2}\right) \tag{19}$$

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where y_0 represents the gravity anomaly value got by the gravimeter at the initial moment of filters. σ_y denotes the standard deviation of the initial gravity anomaly value which is unknown in general. Therefore, the value can be set according to the degree of dependence on the initial gravity anomaly value. When the initial gravity anomaly value is more important, the smaller value can be taken. Otherwise, the larger value can be taken.

In (14)

$$p\left(x_k|x_{k-1}^{(i)}\right) = \frac{1}{(2\pi)^{n/2}|\mathbf{Q}_{k-1}|^{1/2}}\exp\left(-\frac{1}{2}\left(x_k-\left(x_{k-1}^{(i)}+u_{k-1,k}\right)\right)^T \times \mathbf{Q}_{k-1}^{-1}\left(x_k-\left(x_{k-1}^{(i)}+u_{k-1,k}\right)\right)\right) \tag{20}$$

where $\mathbf{Q}_{k-1}$ represents the process noise covariance matrix at time $k-1$, which is also unknown. In this article, it needs to be comprehensively evaluated according to the matching period, INS parameters, and the operating state of the underwater vehicle.

In (15)

$$p_z\left(z_k|x_k^{(i)}\right) = \frac{1}{(2\pi)^{n/2}|\mathbf{R}_k|^{1/2}}\exp\left(-\frac{1}{2}\left(z_k-\left[h\left(x_k^{(i)}\right)\quad q\left(x_k^{(i)}\right)\right]\right)^T \times \mathbf{R}_k^{-1}\left(z_k-\left[h\left(x_k^{(i)}\right)\quad q\left(x_k^{(i)}\right)\right]\right)\right) \tag{21}$$

where $\mathbf{R}_k = ([\begin{smallmatrix}\sigma_{y,k} & 0\\ 0 & \sigma_{r,k}\end{smallmatrix}])^2$. Similar to the definition of σ_y, $\sigma_{y,k}$ and $\sigma_{r,k}$ represent the degree of trust in the gravity anomaly measurement and direction measurement at time k, respectively. The calculation method of $q(x_k^{(i)})$ is shown as

$$q\left(x_k^{(i)}\right) = \arctan\left(\frac{x_k^{(i)(2)}-\hat{x}_{k-1}(2)}{x_k^{(i)(1)}-\hat{x}_{k-1}(1)}\cos\left(x_{k-1}^{(i)(1)}\cdot\pi/180\right)\right) \tag{22}$$

if the unit of position is latitude and longitude. In (22), $x_k^{(i)(1)}$ represents the first element of $x_k^{(i)}$, and $\hat{x}_{k-1}$ represents the posterior estimation at time $k-1$.

Remark 1: It can be seen from (21) that the direction measurement further constrains the particles compared to only gravity measurement. In this article, the direction measurement is taken as a virtual measurement value so that the improved PF is identical to the standard PF in form.

D. Robustness of Improved PF in INS

Due to the inevitable occurrence of outliers in measurement, the filter algorithm is generally optimized to deal with outliers. For the application background of this article, there is a following theorem.

Theorem 1: Under the condition that the initial position is relatively accurate, the improved PF algorithm proposed in this article can deal with the measured outliers and has certain robustness.

Proof: The more accurate position point calculated in Section II is used as the initial point of the improved PF, so the covariance matrix of the initial position of the filter will not be very large. Most of the initial particles are near the real position. Each particle is numerically close to the corresponding gravity anomaly in the gravity background map.

Therefore, when the filter reaches time k, the corresponding covariance matrix will not be very large, and the gravity anomaly value corresponding to each particle in the gravity background map is also relatively close. In improved PF, the likelihood function of the ith particle $x_k^{(i)}$ is $p_z(z_k|x_k^{(i)})$. It can be transformed as follows:

$$p_z\left(z_k|x_k^{(i)}\right) = \frac{1}{(2\pi)^{n/2}|\mathbf{R}_k|^{1/2}}\exp\left(-\frac{1}{2}\left(z_k-\left[h\left(x_k^{(i)}\right)\quad q\left(x_k^{(i)}\right)\right]\right)^T \times \mathbf{R}_k^{-1}\left(z_k-\left[h\left(x_k^{(i)}\right)\quad q\left(x_k^{(i)}\right)\right]\right)\right) \tag{23}$$

$$= \frac{1}{(2\pi)^{n/2}|\mathbf{R}_k|^{1/2}}\exp\left(-\frac{1}{2}\left(y_k-h\left(x_k^{(i)}\right)\right)^T\sigma_{y,k}^{-1}\left(y_k-h\left(x_k^{(i)}\right)\right)\right) - \frac{1}{2}\left(r_k-q\left(x_k^{(i)}\right)\right)^T\sigma_{r,k}^{-1}\left(r_k-q\left(x_k^{(i)}\right)\right)\Bigg) \tag{24}$$

$$= \frac{1}{(2\pi)^{n/2}|\mathbf{R}_k|^{1/2}}\exp\left(-\frac{1}{2}\left(y_k-h\left(x_k^{(i)}\right)\right)^T\sigma_{y,k}^{-1}\left(y_k-h\left(x_k^{(i)}\right)\right)\right) \times \exp\left(-\frac{1}{2}\left(r_k-q\left(x_k^{(i)}\right)\right)^T\sigma_{r,k}^{-1}\left(r_k-q\left(x_k^{(i)}\right)\right)\right) \tag{25}$$

$$= \frac{1}{(2\pi)^{n/2}|\mathbf{R}_k|^{1/2}}O_{y,k}^{(i)}O_{r,k}^{(i)}. \tag{26}$$

When there are outliers in the gravity anomaly measurement values, for all particles, $y_k - h(x_k^{(i)})$ has a large value, and all $O_{y,k}^{(i)}$ have a small value with little difference. $p_z(z_k|x_k^{(i)})$ is mainly decided by $O_{r,k}^{(i)}$ and measured outliers do not have much effect on weight. Therefore, the measured outliers will not cause large deviations. Theorem 1 is proven. ■

Obviously, the factor that resists outliers is mainly $\sigma_{y,k}$. The smaller $\sigma_{y,k}$ induces a stronger ability to resist outliers, but the accuracy of gravity anomaly measurements is required to be high. The larger the $\sigma_{y,k}$ reduces the weaker ability to resist outliers, the lower the accuracy requirements for gravity anomaly measurements.

E. Computational Complexity

In this section, the computational complexity (CC) of the improved PF is analyzed. We use c_1, c_2, c_3, and c_4 to represent the complexity of generating Gaussian distributed random numbers, computing the Gaussian likelihood function, resampling of particles, and multiplication of real numbers. Compared with the standard PF algorithm, the improved PF improves the measurement dimension from 1 to 2. For one iteration, the CC of (15) is $2c_2N$ with the improved PF, while c_2N with the standard PF. The remaining steps are of the same complexity. The CC of (14) is $2c_1N$. The CC of (16) is c_4N. The CC of (17) is $2c_3N$. The CC of (18) is c_4N. Ignore the CC of addition, the CC of the proposed improved PF is

$$\mathrm{CC}_i(N) = 2N(c_1+c_2+c_3+c_4) \tag{27}$$

while the CC of the standard PF is

$$\mathrm{CC}(N) = 2N(c_1+c_2+c_3+c_4) - Nc_2. \tag{28}$$

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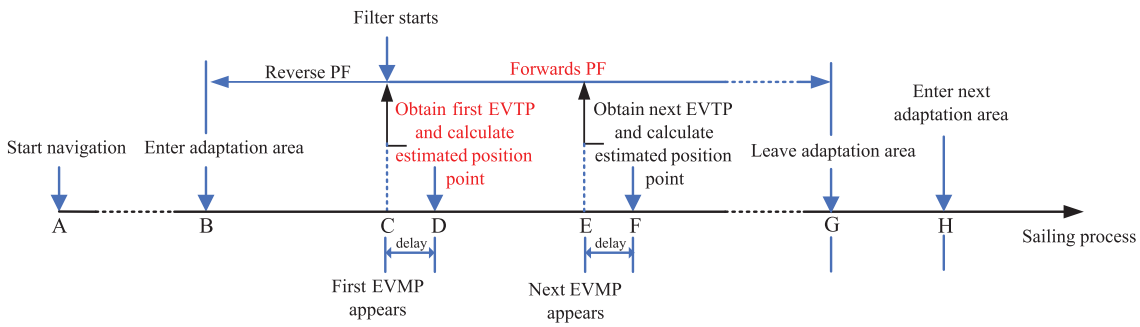

Fig. 7. Gravity matching process in adaptation area.

Due to the addition of direction measurement, the computational burden is increased. The difference and analysis in computation time will be given in Section V.

IV. Improved PF Based on Gravity Measurement Feature

According to the above analysis of gravity measurement features and the improvement of PF, this article proposes an IPFBGMF algorithm. In accordance with the sailing time sequence, the process is shown in Fig. 7.

The underwater vehicle starts to sail, and only INS works at time A. The error increases with time. When entering the adaptation area at time B, INS already has a certain initial position error, the value of which is unknown, but generally does not diverge seriously. During the B∼D time period, the underwater vehicle is maneuvering. At time D, EVTP and EVMP are determined according to the content in Section II. There will be a certain time delay in C∼D. The estimated position point at time C determined at time D is used as the initial value of the improved PF, and time C is the starting point of the filter. At this time, the position estimation can be performed forward or reverse by improved PF. If there are multiple EVMPs in the adaptation area, the obtained new estimated position point is regarded as the measured value for PF. When the underwater vehicle exits the adaptation area at time G, the improved PF stops until it enters the next adaptation area at H time.

V. Semiphysical Simulation and Analysis

This section will verify the effectiveness of the proposed IPFBGMF algorithm. The key factors affecting the accuracy of the IPFBGMF algorithm is the size of the algorithm parameters ($\sigma_{y,k}$, $\sigma_{r,k}$, etc.). This article will use the control variable method to simulate and analyze them.

In semiphysical simulation, the INS path is generated by a dual-axis rotational INS as shown in Fig. 8. The MATLAB codes of all algorithms are carried out with a computer with AMD Ryzen 7 4800H CPU running @ 2.90 GHz. The simulation does not start from the starting point of navigation, but from entering the adaptation area. At this time, INS has a certain initial position error. Some parameters are shown as Table I.

The size of the simulation area in this section is $3° \times 3°$, and its 2-D and 3-D gravity distribution are shown in Fig. 9.

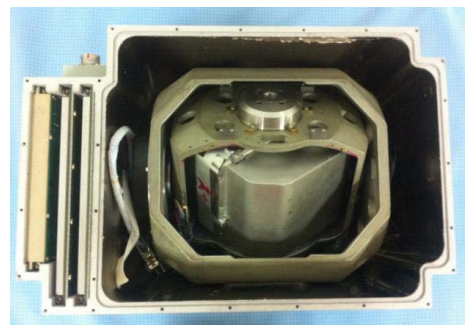

Fig. 8. Dual-axis rotational INS.

TABLE I
Some Simulation Parameters

Parameters	Value
Initial position[latitude, longitude, height]	$[33.7°, 135.1°, 0m]$
Initial velocity[east, north, up]	$[3m/s, 3m/s, 0m/s]$
Initial attitude[pitch, roll, heading]	$[0°, 0°, 45°]$
Gyro constant drift	$0.01°/h$
Gyro random error	$0.001°/h$
Accelerometer bias	$100\mu g$
Accelerometer random error	$50\mu g$
Gravity background map resolution	$1'$

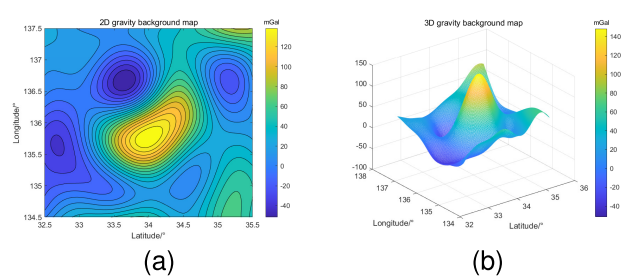

Fig. 9. Gravity background map in simulation. (a) 2-D. (b) 3-D.

A. Simulation on Estimated Position Point

In order to evaluate the accuracy of the method proposed in Section II to find the estimated position point, this section conducts simulations under the conditions of different gravity anomaly measurement accuracy and different INS output heading angle accuracy. The path used for verification is shown in Fig. 10. $N = 11$ in (1).

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TABLE II
ACCURACY OF ESTIMATED POSITION POINT UNDER VARIOUS CONDITIONS

	No noise	Gravity measurement noise standard deviation (mGal)			INS heading angle standard deviation (′)		
		1	2	3	0.05	0.01	0.15
Latitude error(′)	6.3031e-4	0.2740	0.5658	0.7697	9.9076e-4	1.8628e-3	2.7588e-3
Longitude error(′)	4.5631e-3	0.0412	0.0792	0.1069	7.2241e-3	1.2920e-2	1.8960e-2

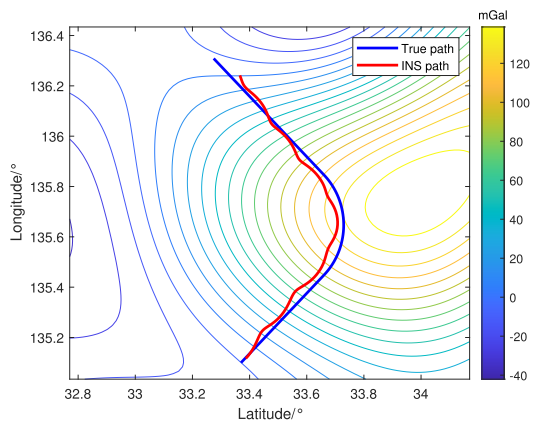

Fig. 10. Path used for verification.

For each gravity anomaly measurement and heading angle, the root mean square error (RMSE) is calculated through 500 Monte Carlo simulations

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{N_m}\left(\hat{x}_i - x_i\right)^2}{N_m}} \tag{29}$$

where N_m is the times of Monte Carlo simulations, $\hat{x}_i$ is the ith estimated value and x_i is ith true value. The simulation result is shown in Table II.

As can be seen from the above Table II, the accuracy of the estimated position point is affected by the accuracy of gravity anomaly measurement and INS output heading angle. As the accuracy decreases, so does the accuracy of the estimated position point. Compared with the heading angle accuracy, the gravity anomaly measurement accuracy has a greater impact. Taking the latitude error as an example, compared with the latitude error of $6.3031 \times 10^{-4\prime}$ in the case of no gravity anomaly measurement error, the latitude error can reach $0.7697'$ when the error standard deviation is 3 mGal. The error is only $2.7588 \times 10^{-3\prime}$ when INS heading angle error standard deviation is 0.15°. The reason is that: the position estimation method based on the gravity measurement feature needs to use the measured gravity anomaly value to generate the point set corresponding to the contour line. When there is no measurement error, the point set contains the true value of the current position. While the measurement error occurs, it means that the sought contour point set is not the real contour point set, and the real point is not included in the point set. Even if the heading angle has no error, positioning error will definitely occur. Compared with the sparse contour line distribution, when the contour line distribution is dense, the gravity anomaly measurement error will make the point set of the real contour line and the measured contour line close to each other. The position error obtained by calculation is small, which is consistent with the characteristics of the adaptation area. Generally speaking, the heading angle error of INS is smaller than the angle difference between two adjacent points on the contour line, so even if there is a certain heading angle error, it may not cause too much deviation from the points searched on the contour line.

TABLE III
SIMULATION PARAMETERS

Parameters	Value
Matching interval	$140s$
Initial position error (latitude, longitude)	$[-0.0014\prime, 0.1345\prime]$
The number of particles	500
$\sigma_{r,k}$	$1°$
Q_k	$\left(\begin{bmatrix} 0.1 & 0 \\ 0 & 0.1 \end{bmatrix}'\right)^2$
P_0	$\left(\begin{bmatrix} 0.3 & 0 \\ 0 & 0.3 \end{bmatrix}'\right)^2$

In conclusion, in order to improve the accuracy of estimated position points, the research focus can be put on improving the accuracy of gravity anomaly measurement. In addition, the path planning in the adaptation area can also plan the path meeting the conditions to reduce the influence of the measurement error on the positioning accuracy.

B. Parameter Simulation on IPFBGMF

Since the IPFBGMF algorithm is affected by the algorithm parameters $(\sigma_{y,k}, \sigma_{r,k})$ and other factors, this section aims to verify the effectiveness of the proposed algorithm in different situations.

1) Effect of $\sigma_{y,k}$ on IPFBGMF: $\sigma_{y,k}$ mainly reflects the importance of gravity anomaly measurement at time k. The smaller $\sigma_{y,k}$ is, it indicates that the algorithm trusts the gravity anomaly measurement more. The control variable method is adopted in this section of the simulation, and all parameters except $\sigma_{y,k}$ remain unchanged. The value of other parameters is shown in Table III, and there is no gravity anomaly measurement error during simulation.

When $\sigma_{y,k}$ values are 0.25, 0.50, 0.75, 1.00, 1.25, and 1.50 mGal, respectively, forward and reverse estimation is carried out at EVTP through the IPFBGMF algorithm. EVTP is 4.628 h of navigation in the adaptation area. The forward navigation in this section refers to the filter from the EVTP to the back according to the time sequence, which is real-time. Reverse navigation refers to smoothing the past path, which is not real-time. The simulation results of latitude error, longitude

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TABLE IV
$\sigma_{y,k}$ SIMULATION STATISTICAL RESULTS WITHOUT GRAVITY MEASUREMENT ERROR

$\sigma_{y,k}$ (mGal)	Reverse navigation						Forward navigation					
	Latitude Error (′)		Longitude error (′)		Position error (nmile)		Latitude Error (′)		Longitude error (′)		Position error (nmile)	
	Mean	STD	Mean	STD	Mean	STD	Mean	STD	Mean	STD	Mean	STD
0.25	0.0545	0.0174	0.0823	0.0896	0.0978	0.0628	0.097	0.0647	0.056	0.0572	0.1177	0.0646
0.50	0.1098	0.0473	0.1780	0.1242	0.2245	0.1213	0.1217	0.0841	0.1242	0.1382	0.1867	0.1043
0.75	0.1749	0.0849	0.3790	0.2533	0.3918	0.1681	0.1198	0.0792	0.1617	0.1495	0.2133	0.0931
1.00	0.2013	0.1017	0.5004	0.2694	0.4900	0.1859	0.1530	0.1209	0.3530	0.2401	0.3724	0.1479
1.25	0.2260	0.1376	0.6941	0.3198	0.6198	0.2208	0.1283	0.0937	0.3847	0.2209	0.3722	0.1531
1.50	0.2108	0.1375	0.8893	0.3260	0.7927	0.2457	0.1965	0.1826	0.4110	0.3329	0.4119	0.3048

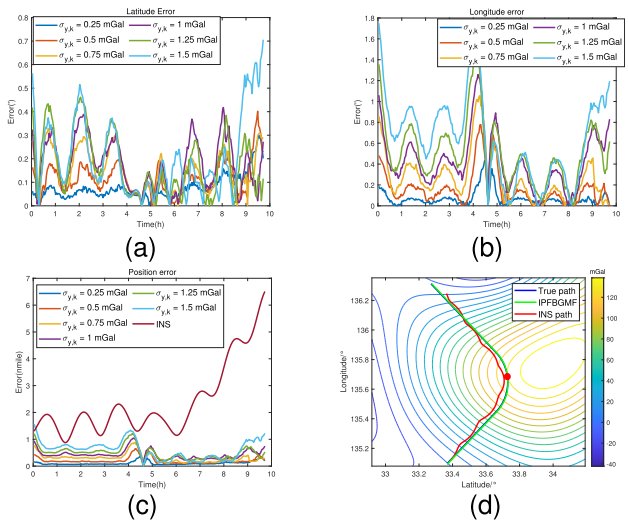

Fig. 11. Error and path under different $\sigma_{y,k}$. (a) Latitude. (b) Longitude. (c) Position. (d) Matching path. (The unit of longitude and latitude error is ′, and the unit of position error is nautical miles.)

error, position error, and the matching path when the value is 0.25 mGal are shown in Fig. 11 and Table IV, in which STD represents the standard deviation.

The algorithm starts to perform forward and reverse navigation methods at 4.628 h. In Fig. 11, there will be a brief error overshoot when the navigation starts. This is not caused by the failure of the IPFBGMF algorithm to converge, but because the path enters an area of poor adaptation. The specific explanation is as follows: As shown in Fig. 12(d), the red point is the estimated position point obtained in Section II, and it is also the initial position point of the IPFBGMF algorithm. It can be seen that through the analysis of the adaptability in [5], according to the real path direction and contour distribution, the area near the estimated position point has poor adaptability in the path direction. The matching method IPFBGMF proposed in this article is also affected by adaptability. According to Fig. 11(c), it can be seen that even in areas with poor adaptability, the positioning error is still no more than 2 nmi at most, which is equivalent to the inertial navigation error. And then the positioning error in the adaptation area will be smaller, which also shows the effectiveness of the algorithm proposed in this article.

The simulation is completed without the gravity measurement error, so the gravity anomaly measurement value is accurate. Theoretically $\sigma_{y,k}$ should be 0. In fact, since in (21) $\sigma_{y,k}$ is in the denominator position, it cannot be set to 0. When the value of $\sigma_{y,k}$ is 0.25 and 0.5 mGal, the latitude, longitude, and position errors are significantly smaller than other values, and the changes in the errors are relatively stable. As $\sigma_{y,k}$ gets bigger and bigger, the error also gets bigger and bigger, and the error changes more and more drastically. Even in the case of 1.5 mGal, the maximum error of reverse navigation and forward navigation is only about 1.5 nmi, which is much smaller than the position error of INS. From Table IV, when $\sigma_{y,k}$ is greater than 0.75 mGal, the forward and reverse latitude, longitude, and position errors change slowly. Therefore, it can be concluded that when $\sigma_{y,k}$ is set to be the same or similar to the standard deviation of the actual gravity measurement error, the IPFBGMF algorithm has high accuracy, and the algorithm error is greatly affected by $\sigma_{y,k}$. The farther the value of $\sigma_{y,k}$ is from the true standard deviation, the greater the error of the IPFBGMF algorithm, and the less affected by $\sigma_{y,k}$ it is.

2) Effect of $\sigma_{r,k}$ on IPFBGMF:

$\sigma_{r,k}$ mainly reflects the importance attached to the direction measurement at time k. The smaller $\sigma_{r,k}$ is, the more the algorithm trusts the direction measurement. This section adopts the control variable method. Other parameters are unchanged except $\sigma_{r,k}$. $\sigma_{y,k}$ is 0.5 mGal, and the other parameters are shown in Table III. There is no gravity measurement error during the simulation.

The values of $\sigma_{r,k}$ are 0.5°, 10°, 20°, 40°, 60°, 80°, and infinity, respectively. The infinity is that the IPFBGMF algorithm does not consider the direction measurement. It is worth explaining that although this article uses the INS heading angle as the direction measurement, $\sigma_{r,k}$ is not the standard deviation of the heading angle error, but the standard deviation of the deviation between the direction of the particle $x_k^{(i)}$ and the estimated position $\hat{x}_{k-1}$ and the direction measurement. The simulation conditions are the same as those in Section V-B.1. The error results and the matching result when $\sigma_{r,k}$ is 0.5° are shown in Fig. 12 and Table V.

It can be seen from Fig. 12 that when $\sigma_{r,k}$ takes infinity, that is, the algorithm only has gravity anomaly measurement, and the position error is larger than when $\sigma_{r,k}$ takes other values. With the extension of time, its positioning error increases with the increase of INS error. Therefore, adding direction measurement can significantly improve the accuracy of the PF algorithm. In addition, when the value of $\sigma_{r,k}$ is from 0.5° to 80°, the position accuracy has a less obvious decline, but it is significantly higher than when it is set to infinity.

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TABLE V
$\sigma_{r,k}$ SIMULATION STATISTICAL RESULTS WITHOUT GRAVITY MEASUREMENT ERROR

$\sigma_{r,k}$ (°)	Reverse navigation						Forward navigation					
	Latitude Error (′)		Longitude error (′)		Position error (nmile)		Latitude Error (′)		Longitude error (′)		Position error (nmile)	
	Mean	STD	Mean	STD	Mean	STD	Mean	STD	Mean	STD	Mean	STD
0.5	0.0971	0.0564	0.1803	0.1670	0.1998	0.1205	0.1389	0.0957	0.1222	0.1259	0.1954	0.1078
10	0.1697	0.0558	0.2679	0.1987	0.3038	0.1289	0.1422	0.0805	0.1179	0.1222	0.1980	0.0860
20	0.2051	0.0646	0.2987	0.2016	0.3475	0.1239	0.1176	0.0871	0.1017	0.1023	0.1632	0.0958
40	0.2153	0.0524	0.2871	0.1838	0.3426	0.1105	0.1730	0.1036	0.1072	0.0938	0.2127	0.0972
60	0.2581	0.0588	0.3237	0.1930	0.3935	0.1164	0.1996	0.1257	0.1206	0.1040	0.2428	0.1191
80	0.2451	0.0556	0.3082	0.1937	0.5736	0.1791	0.2430	0.1647	0.1291	0.1149	0.3055	0.1179
Inf	0.3322	0.2565	0.3504	0.2532	0.4494	0.3225	1.5900	1.4876	0.5863	0.3371	1.7623	1.3965

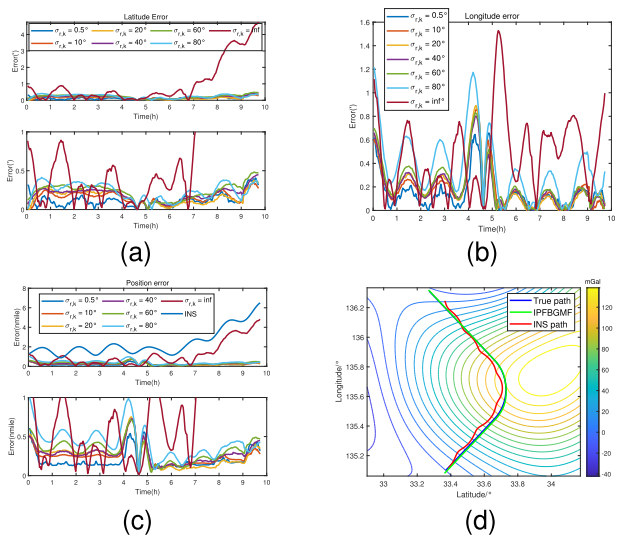

Fig. 12. Error and path under different $\sigma_{r,k}$. (a) Latitude. (b) Longitude. (c) Position. (d) Matching path.

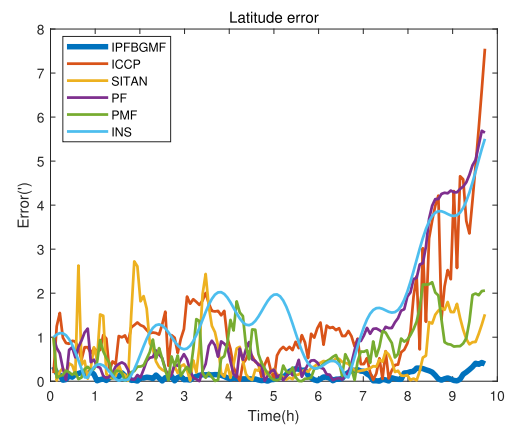

Fig. 13. Latitude error comparison under different matching algorithms with gravity measurement error.

In Fig. 12(c), the reverse navigation error increases as $\sigma_{r,k}$ increases. In forward navigation, when $\sigma_{r,k}$ is less than 40°, there is little difference in position error. When $\sigma_{r,k}$ is greater than 40°, it increases as the value of $\sigma_{r,k}$ increases. But, in general, no matter what the value of $\sigma_{r,k}$ is, the position error is less than 0.5 nmi. The reason for the small difference in position error is that: in the PF, (15) is performed on each particle. Due to the large number and wide distribution of particles, the corresponding direction measurement error of each particle

$$\varepsilon_k^{(i)} = z_k - \eta\left(x_k^{(i)}\right) \tag{30}$$

is quite different. Therefore, even with large values of $\sigma_{r,k}$, formula (15) can still give high weights to particles which are near the right direction measurement. Therefore, through this section of the simulation, it can be concluded that adding direction measurement can effectively improve the positioning accuracy; within a suitable range, the value of $\sigma_{r,k}$ will not have an excessive impact on the positioning accuracy.

C. Comparison With Other Gravity Matching Algorithms

This section aims to compare with the traditional gravity matching algorithms to verify the effectiveness and advantages of the IPFBGMF algorithm proposed in this article. The comparison algorithms are ICCP, SITAN, PF, TERCOM, and PMF.

There are inevitably errors in the gravity anomaly measurement. In this section, the above algorithms are simulated and compared under the condition of considering the gravity measurement error. The initial position error of INS when entering the adaptation area is 1.303 nmi; the period of the matching algorithm is 250 s; the ICCP algorithm adopts the sliding window method, and the number of matching sequences is 6; in the SITAN, PF, and PMF algorithms, the measurement covariance matrix $\mathbf{R}_k = 1$ mGal; the initial covariance matrix $\mathbf{P}_0$ is diag([1′; 1′; 1.2 m/s; 0.9 m/s; 6′; 6′; 6′; 100 μg; 100 μg; 100 μg; 0.005°/h; 0.005°/h; 0.005°/h]^2) and the process noise covariance matrix $\mathbf{Q}_k$ is diag([50 μg; 50 μg; 50 μg; 0.003°/h; 0.003°/h; 0.003°/h]^2) with initial state $\mathbf{x}_0 = \mathbf{0}_{13\times 1}$ in SITAN; the number of particles in the PF algorithm is 500; the process and measurement error covariance matrix are both diag([0.3′; 0.3′]^2) and diag(2 mGal)^2 in both PF and PMF; the initial position error covariance matrix is diag([1′; 1′]^2); the effective radius error of latitude and longitude in the TERCOM algorithm is 1′; Mean absolute differences (MADs) is applied in TERCOM; in the IPFBGMF algorithm, $\sigma_{y,k}$ takes 0.25 mGal and $\sigma_{r,k}$ takes 1°. The gravity measurement error is set to obey a Gaussian distribution with a mean of 0 and a standard deviation of 1 mGal. Simulation results are shown in Figs. 13–15 and Table VI. The last column of Table VI represents the calculation time of once gravity matching process.

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TABLE VI
COMPARISON OF MATCHING ALGORITHMS WITH GRAVITY MEASUREMENT ERROR

Navigation method	Latitude Error (′)		Longitude error (′)		Position error (nmile)		Calculation Time(s)
	Mean	STD	Mean	STD	Mean	STD	
IPFBGMF	0.1197	0.0964	0.1638	0.1148	0.1866	0.1287	4.9778
ICCP	1.3374	1.2835	0.8354	0.6480	1.6074	1.2750	3.6192
SITAN	0.6074	0.6178	0.4723	0.4625	0.7392	0.7127	1.9582
PF	1.1485	1.4734	0.4838	0.3891	1.3069	1.4206	4.8693
PMF	0.6542	0.5833	0.6267	0.6488	0.9460	0.6597	5.7177
TERCOM	3.7126	2.0839	9.5429	5.9677	9.0962	4.8436	1.8932
INS	1.5211	1.2670	1.9494	1.0687	2.3682	1.3217	-

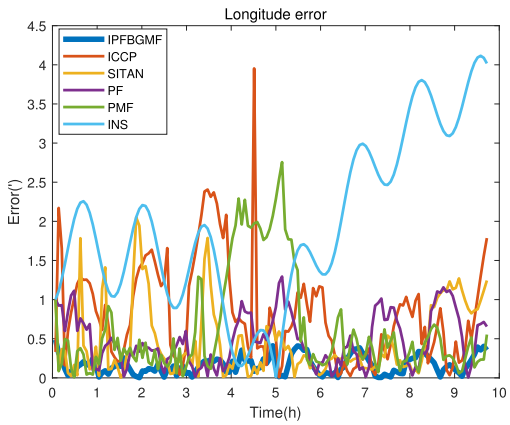

Fig. 14. Longitude error comparison under different matching algorithms with gravity measurement error.

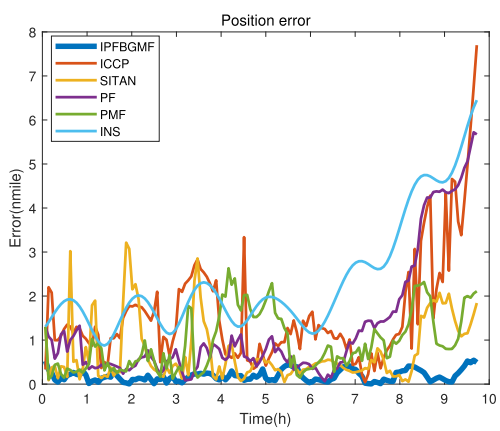

Fig. 15. Position error comparison under different matching algorithms with gravity measurement error.

As shown in Fig. 15 and Table VI, although the proposed algorithm requires more computing time, the time is much less than the period of gravity matching, which shows that real-time requirement can be met. The position error of IPFBGMF is about 0.1866 nmi, which is much smaller than the position error of 2.3682 nmi of INS. Compared with ICCP, SITAN, PF, and PMF, the accuracy of IPFBGMF is improved by 88.37%, 74.76%, 85.72%, and 80.27%. TERCOM algorithm is failed. In the case of existing gravity measurement errors, the IPFBGMF algorithm proposed in this article has the highest accuracy.

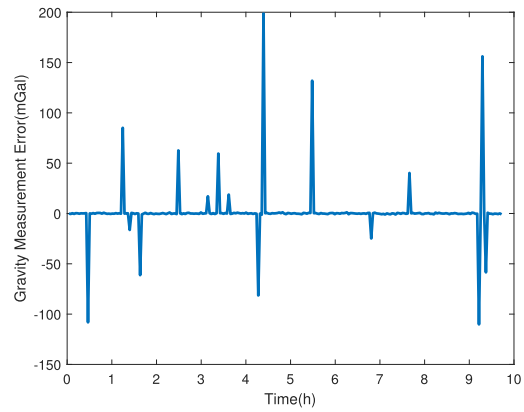

Fig. 16. Gravity measurement errors.

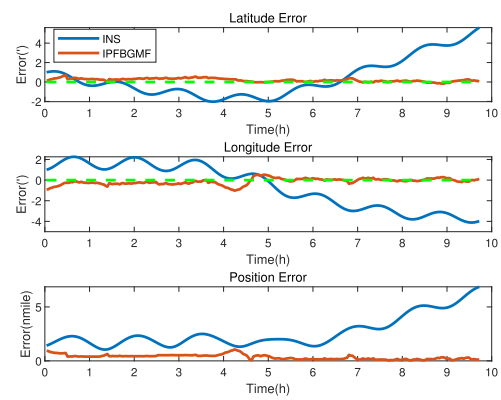

Fig. 17. Matching results with outliers.

In conclusion, regardless of the existence of gravity measurement errors, the IPFBGMF algorithm has the best positioning accuracy.

D. Simulation With Measured Outliers

In order to verify Theorem 1, that is, to verify that the IPFBGMF algorithm has certain robustness, this section is designed to perform simulation verification under the condition that the measurement has outliers. The relevant simulation parameters are the same as those in Section V-C. Without considering other errors such as gravity background map and calculations, the distribution of gravity measurement error υ_k is set as

$$\upsilon_k \sim \begin{cases} N(0,1) & w.p.0.95 \\ N(0,100) & w.p.0.05 \end{cases}. \tag{31}$$

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One of the gravity measurement errors is shown in Fig. 16. The positioning error results of the IPFBGMF algorithm are shown in Fig. 17. Although there are outliers in gravimetric measurement, the IPFBGMF algorithm can still suppress the outlier phenomenon to a certain extent, so Theorem 1 is verified.

VI. Conclusion

The gravity-matching algorithm plays a significant role in GAINS. This article proposes a novel IPFBGMF algorithm. The gravity measurement feature is studied and a precise estimated position point is achieved by doing research on the contour line and INS. The point can be the initial position of PF. What is more, a new concept called direction measurement is proposed and the PF is improved. The robustness of the improved PF is also proved. A gravity-matching strategy is given which combines the above algorithms. Simulations show that the IPFBGMF algorithm has better positioning accuracy than most gravity-matching algorithms and has robustness.

Although IPFBGMF has better results while estimated position point is obtained conditionally that is the underwater vehicle needs to maneuver in some areas. Therefore, in future work, it is necessary to do more research on gravity features and make matching algorithms unconstrained.

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